Dr. Filippo Geraci : Algorithmic game theory

March : Mon 3/3, Thu 6/3, Fri 7/3
16-18

Wed 12/3, Thu 13/3, Fri 14/3
16-18

Indicator variable $X = \begin{cases} 1 & \text{event ON} \quad p \\ 0 & \text{event OFF} \quad 1-p \end{cases}$ 

$E[X] = p = Pr(X=1)$

Buy headphones on the web

n headphones $\underset{\text{worst}}{\underset{\uparrow}{1}}, 2, \dots, \underset{\text{best}}{\underset{\uparrow}{n}}$

example $n=9$ headphones

$\boxed{1}\ \boxed{2}\ \boxed{3} \cdots \boxed{9}$

9 3 4 ... 8 how good is the headphone

cost = 1

3 4 2 5 6 1 7 8 9 cost = 7

1 2 3 4 5 6 7 8 9 cost = n

$x_1 \quad x_2 \quad x_3 \quad \ldots \quad x_9$

5 1 6 9 2 7 4 8 3   no cost = 3

7 5 2 6 4 8 1 9 3   no cost = 3

cost = ..

$+ \ldots$

$$\frac{\quad}{n!} = E(\text{cost})$$

$n!$

$$H = \begin{array}{c} \left[\begin{array}{ccccccccc} \bar{2} & \bar{4} & 3 & \bar{5} & 1 & \bar{7} & \bar{8} & 6 & \bar{9} \end{array}\right] \\ \begin{array}{ccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \end{array} \end{array}$$

$$X_1 = X_2 = X_4 = X_6 = X_7 = X_9 = 1$$

0s the rest

$$X_i = \begin{cases} 1 & \text{if } H[i] \text{ is chosen} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{cost} = \sum_{i=1}^{n} X_i$$

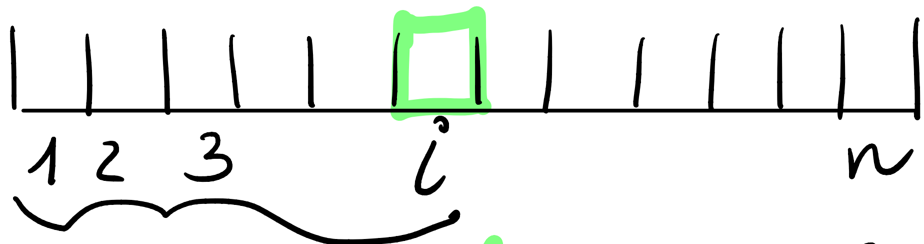
$$E[\text{cost}] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} \underbrace{\Pr(X_i = 1)}$$

linearity of $E[\ ]$

$$= \sum_{i=1}^{n} \frac{1}{i} \sim \ln n$$

Harmonic sum

▶ $\Pr(X_i = 1) = \frac{\cancel{f}\,(i-1)!}{\cancel{f}\,i!} = \frac{1}{i}$

$H =$ 

ON iff $H[i] > H[j]$ for $1 \leq j < i$

f = # extra permutations extending from i to n

Given an array A of n distinct elements, permute them uniformly at random $\to$ SHUFFLING

random$(a,b) \to x \in [a..b]$ uniformly at random

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for i = 1 .. n-1 do:
    J = random(i .. n)
    swap A[i] and A[J]
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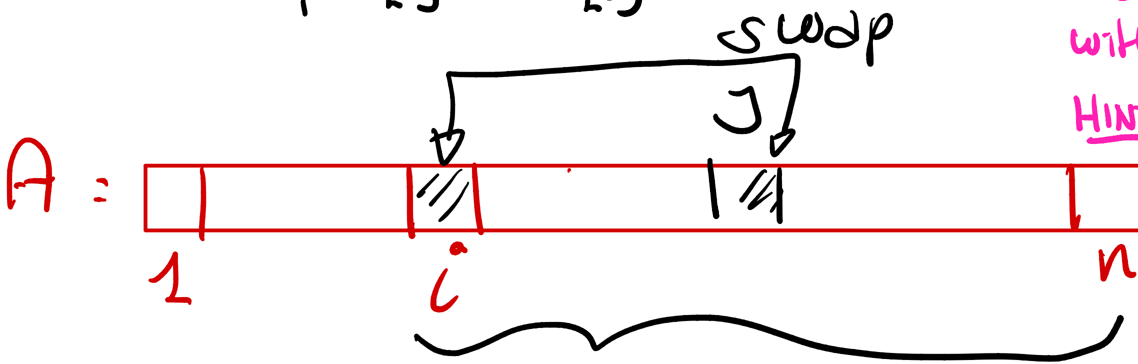


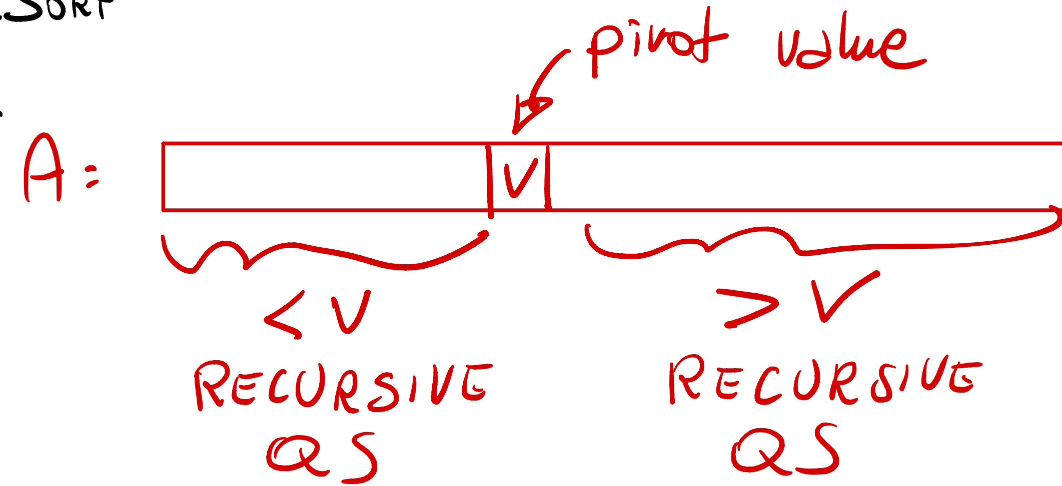
Exercise | Use indicator variable to show that this for loop gives one of the $n!$ permutations with $pr = \frac{1}{n!}$

HINT: By induction on i: $A[1..i]$ is the prefix of $(n-i)!$ permutation with prob. $\frac{(n-i)!}{n!}$

HINT: $\frac{1}{n-i+1} \cdot \frac{(n-i+1)!}{n!}$?

Randomized QuickSORT

QS:



Randomization: choose v uniformly at random among the n elements of A

Randomized algorithm → expected cost ← probability space are all random choices of the pivot position in A (FIXED)

Deterministic algorithm → expected cost
← the probability space are all possible A's

A sorted $z_1 < z_2 < z_3 \cdots < z_n$

(for the sake of discussion)

$$X_{ij} = \begin{cases} 1 & \text{when } z_i \text{ and } z_j \text{ are compared} \\ 0 & \text{otherwise} \end{cases}$$

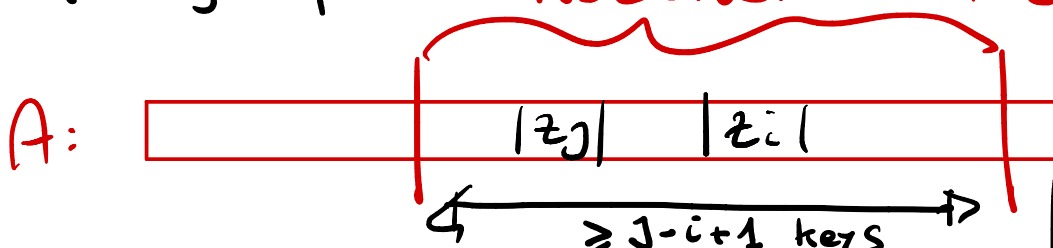
Fact

1] z_i and z_j compared $\Rightarrow$ one of them is a pivot

2] z_i and z_j are compared at most once

$$\sum_{1 \le i < j \le n} X_{ij} = \text{total number of comparisons made by the randomized QS}$$

3] z_i and z_j compared RECURSIVE CALL $\Rightarrow$ $z_i, z_{i+1}, \dots, z_{j-1}, z_j$ are all inside this call



$|I| \ge j-i+1$

expected cost

$$E\left[\sum_{i<j} X_{ij}\right] = \sum_{i<j} E[X_{ij}] = \sum_{i<j} \underbrace{Pr(X_{ij}=1)} = \sum_{i<j} \frac{2}{j-i+1}$$

$$Pr(X_{ij}=1) = Pr\left(\begin{array}{l} z_i, z_j \text{ are in the same call } I \\ \underline{\text{and}}\ z_i \text{ or } z_j \text{ is a randomly chosen pivot in } I \end{array}\right)$$

$$\leq Pr(z_i \text{ or } z_j \text{ is a randomly chosen pivot in } I)$$

$$= \frac{2}{|I|} \leq \frac{2}{j-i+1}$$

$$= \sum_{i<j} \frac{2}{j-i+1} = \sum_{i=1}^{n} \sum_{j=i+1}^{n} \frac{2}{j-i+1} = O(n \lg n)$$

Harmonic sum

$O(\lg n)$

$\blacksquare$